

Multiply and ACcumulate (MAC)

Digital Signal Processing

Digital Signal Processing/Processor - DSP

The same acronym, DSP, refers both to the application domain and to the circuits for its elaboration.

As a matter of facts, all DSP applications (filters, transformations, convolutions, ...) are obtained executing the sum of products:

$$\sum_{k=1}^N x_k y_k$$

For example, the Fourier Transform:

$$F_h = \sum_{k=1}^N f_k e^{-i k_h x_k}$$

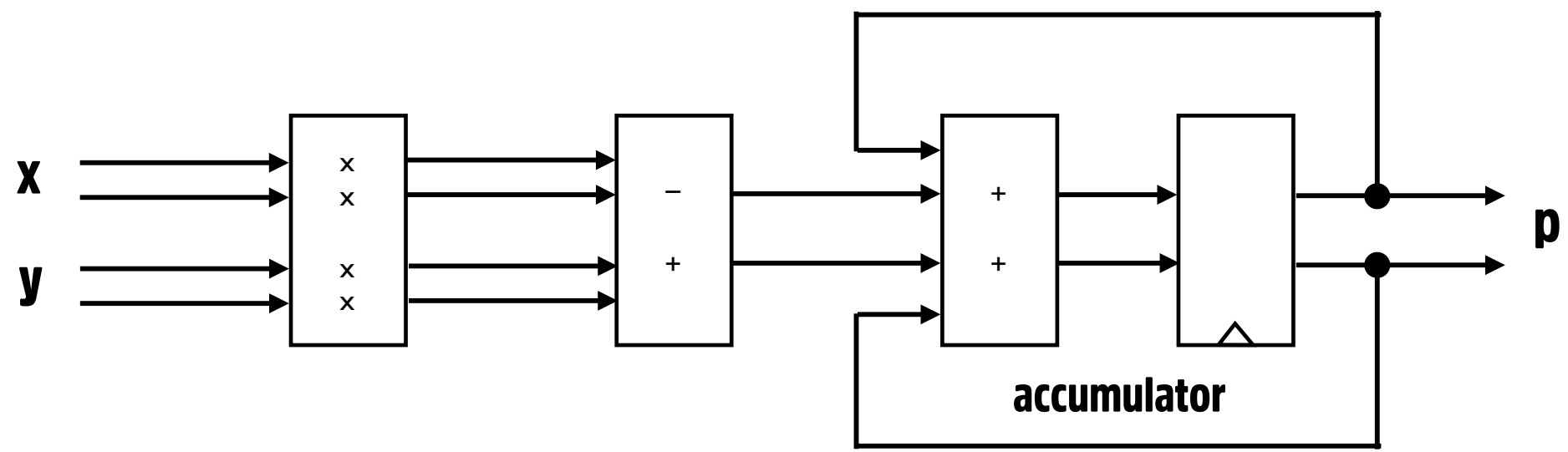
Complex Sum and Product

As it is known, in the complex number domain the sum and product operator act to the complex numbers, say $x = x_r + i x_i$ as follows:

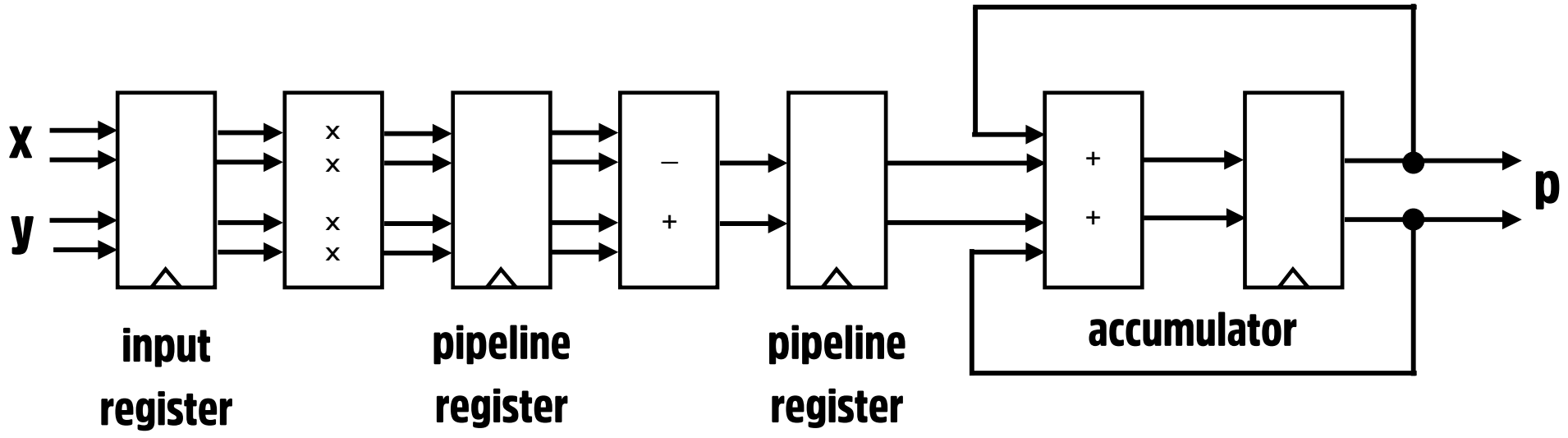
$$s = x + y = (x_r + y_r) + i (x_i + y_i)$$

$$p = x \times y = (x_r y_r - x_i y_i) + i (x_r y_i - x_i y_r)$$

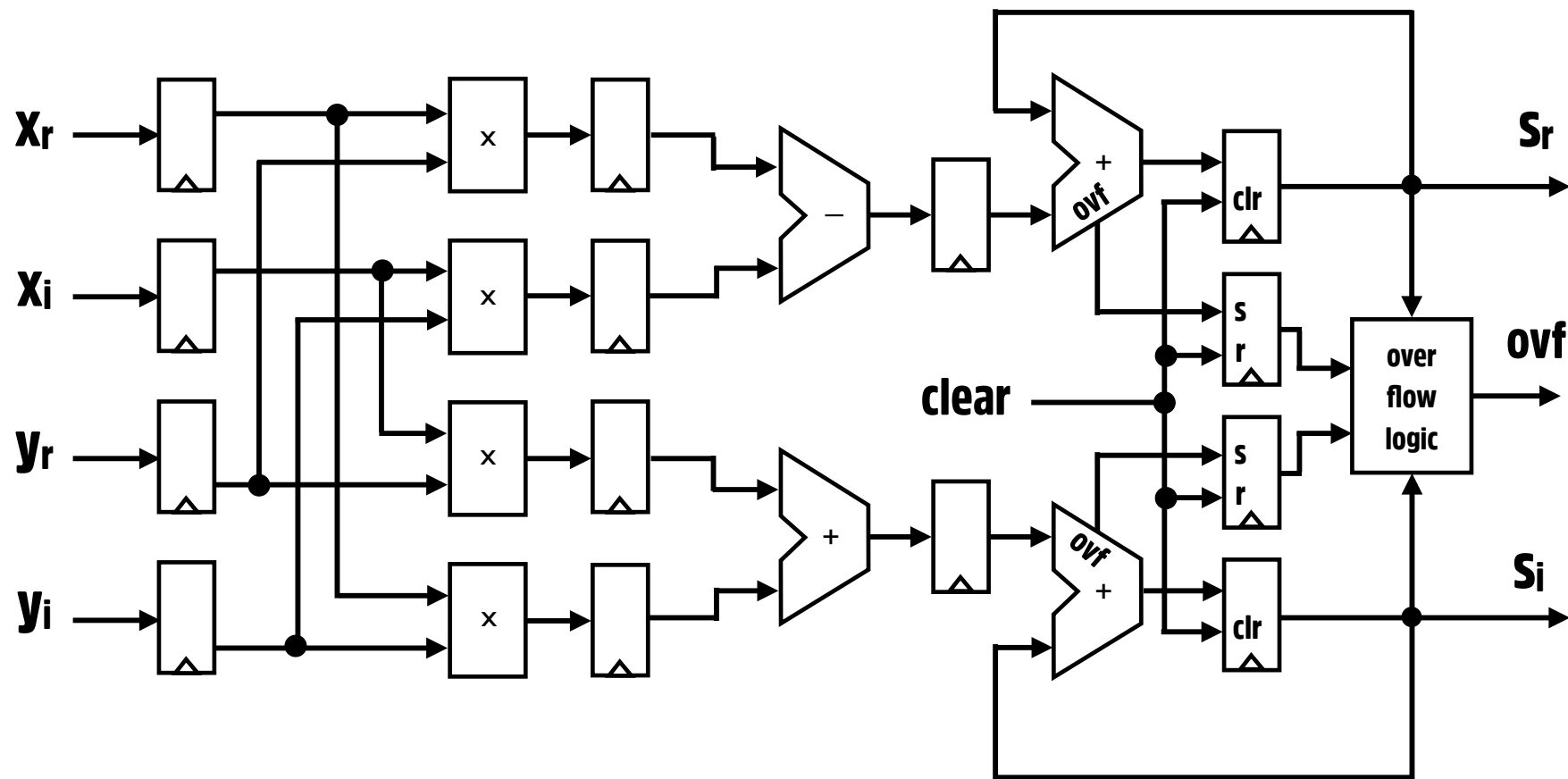
Single Cycle MAC



Pipelined MAC



RTL Structural View



Fixed-point Representation

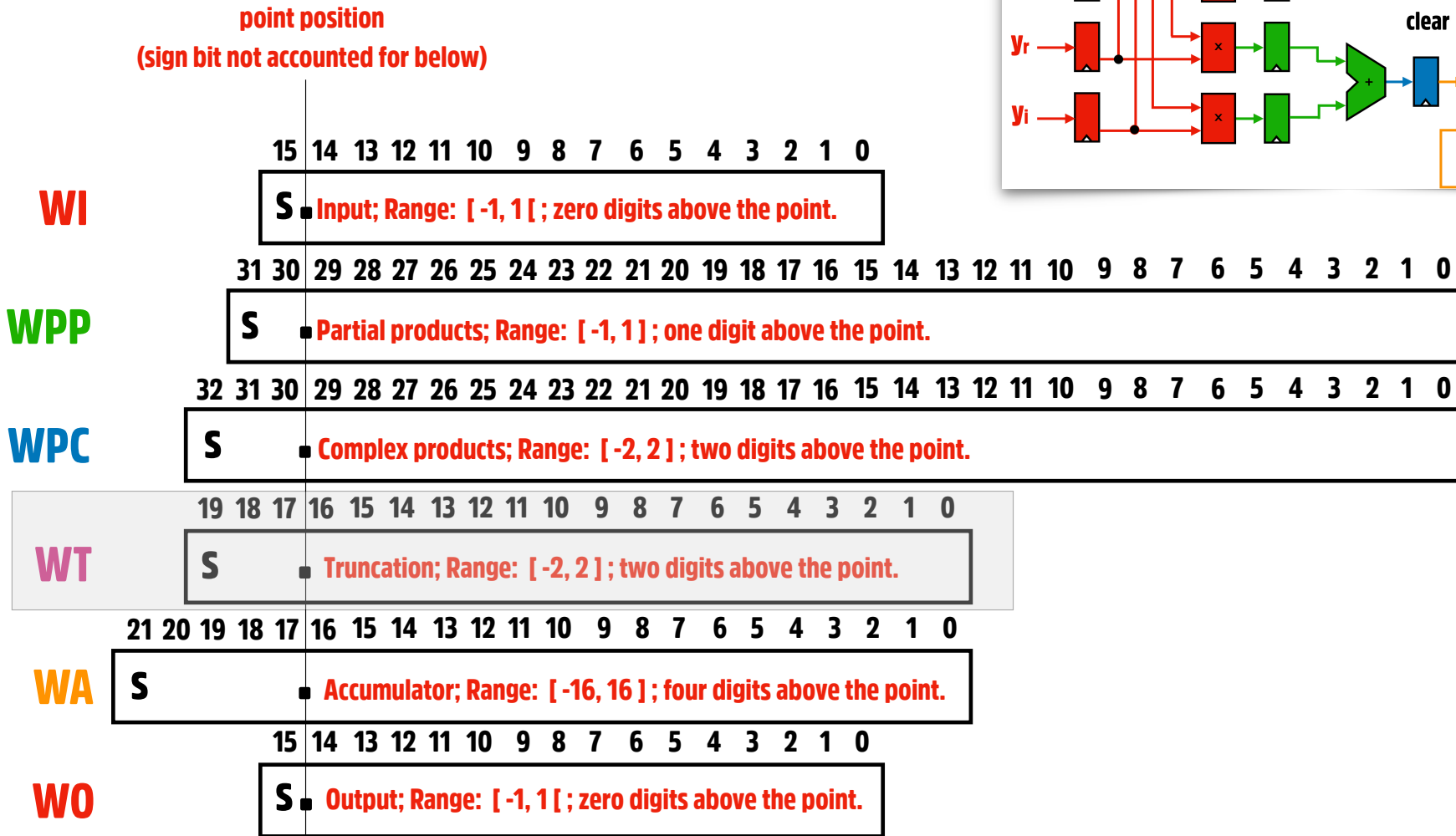
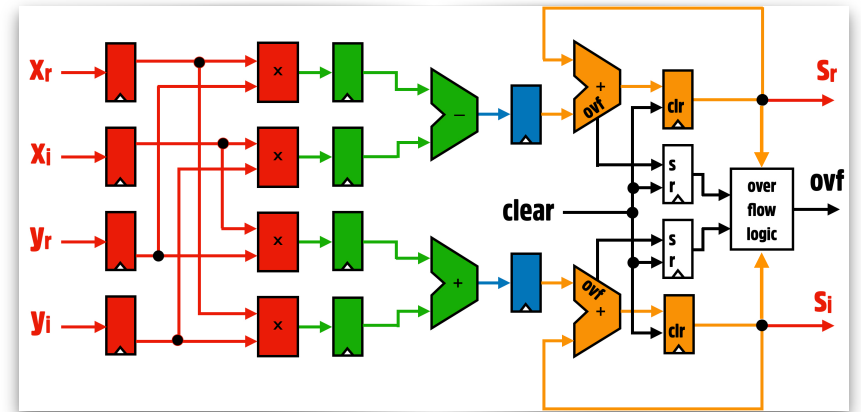
bit index

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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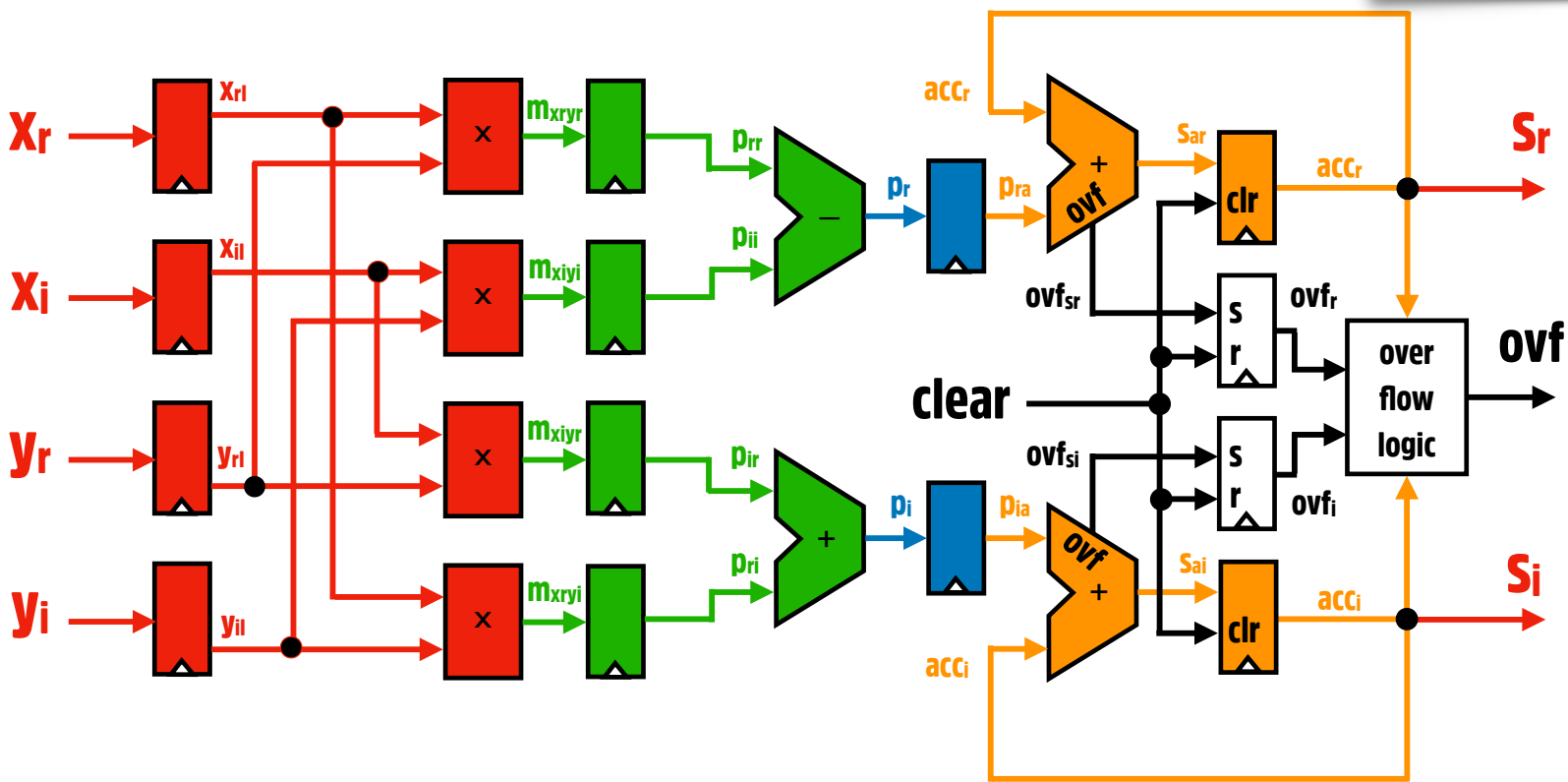
bit weight

-2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-10}	2^{-11}	2^{-12}	2^{-13}	2^{-14}	2^{-15}
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Data Formats of the Datapath



MAC RTL Structural View

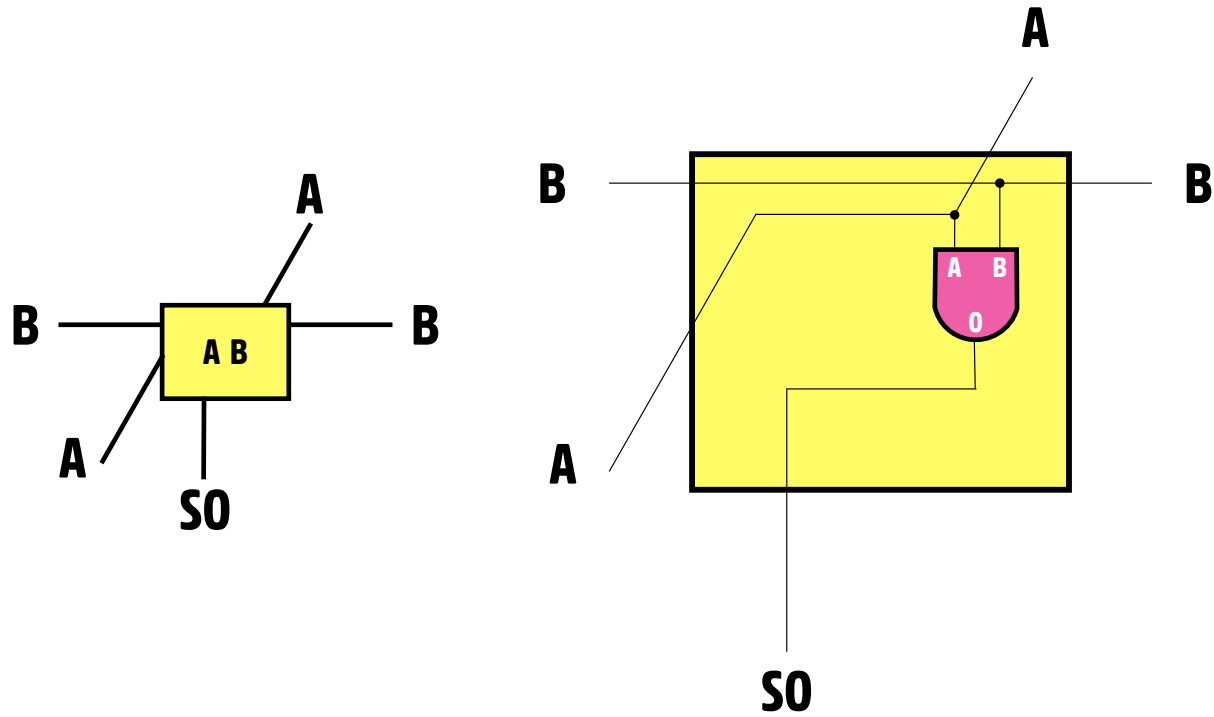


point position		
	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	
WI	S	Input; Range: [-1, 1]; zero digits above the point.
WPP	S	Partial products; Range: [-1, 1]; one digit above the point.
WPC	S	Complex products; Range: [-2, 2]; two digits above the point.
WT	S	Truncation; Range: [-2, 2]; two digits above the point.
WA	S	Accumulator; Range: [-16, 16]; four digits above the point.
WO	S	Output; Range: [-1, 1]; zero digits above the point.

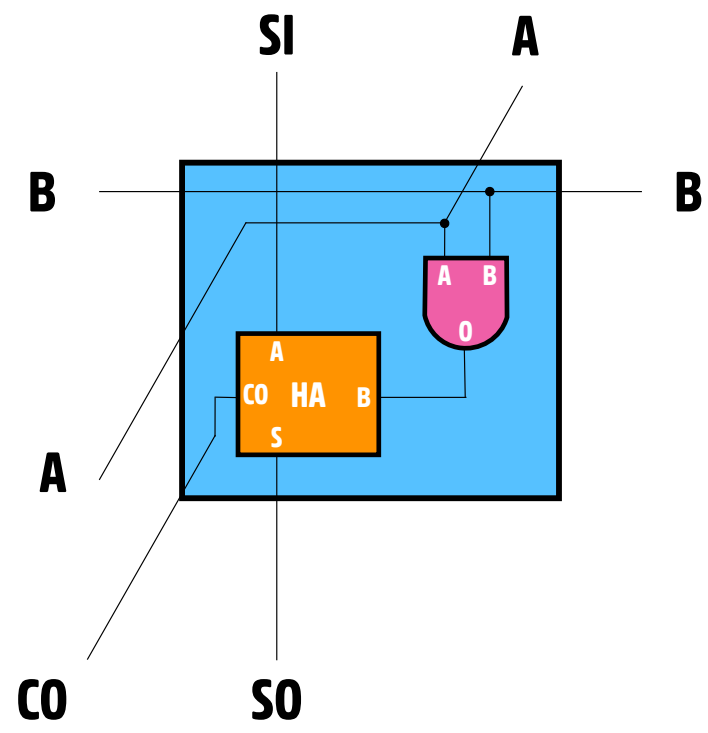
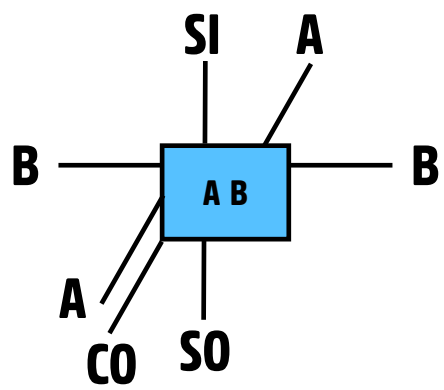
Binary Multiplication

					A₃	A₂	A₁	A₀	×
					B₃	B₂	B₁	B₀	=
				A₃ B₀	A₂ B₀	A₁ B₀	A₀ B₀		
			A₃ B₁	A₂ B₁	A₁ B₁	A₀ B₁			
		A₃ B₂	A₂ B₂	A₁ B₂	A₀ B₂				
	A₃ B₃	A₂ B₃	A₁ B₃	A₀ B₃					
S₇	S₆	S₅	S₄	S₃	S₂	S₁	S₀		

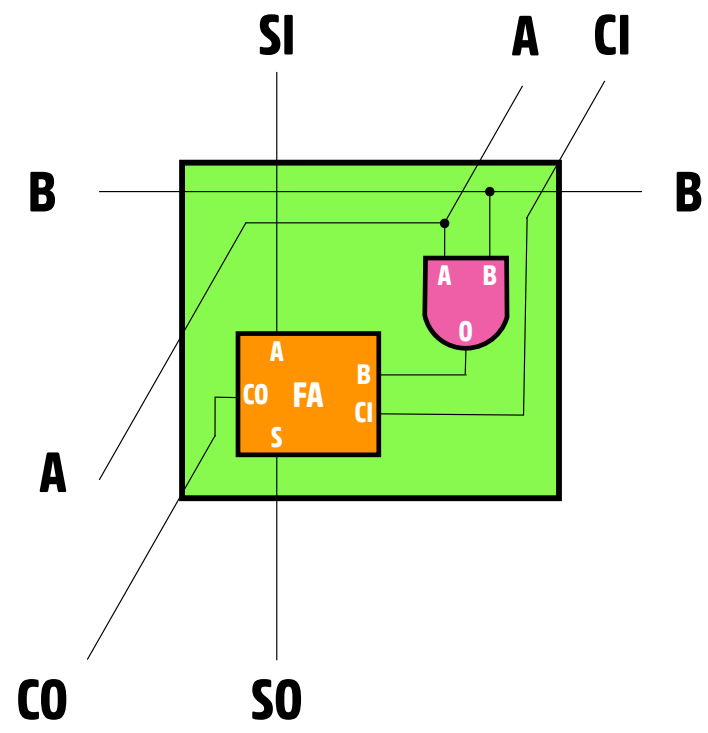
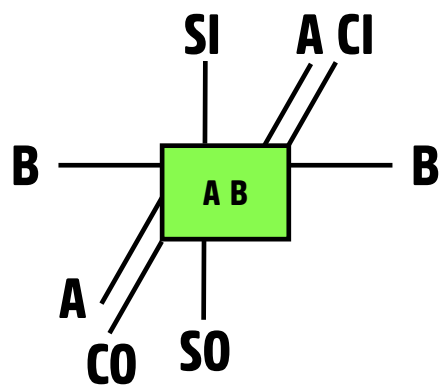
Binary Multiplication - Block #1



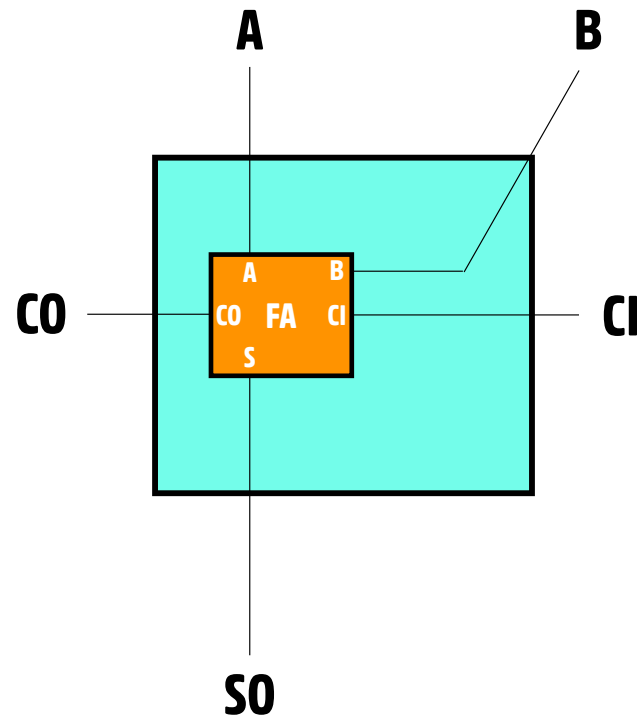
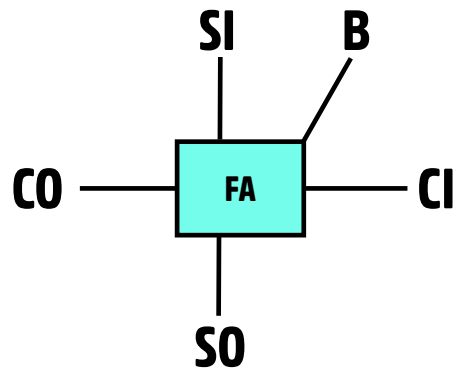
Binary Multiplication - Block #2



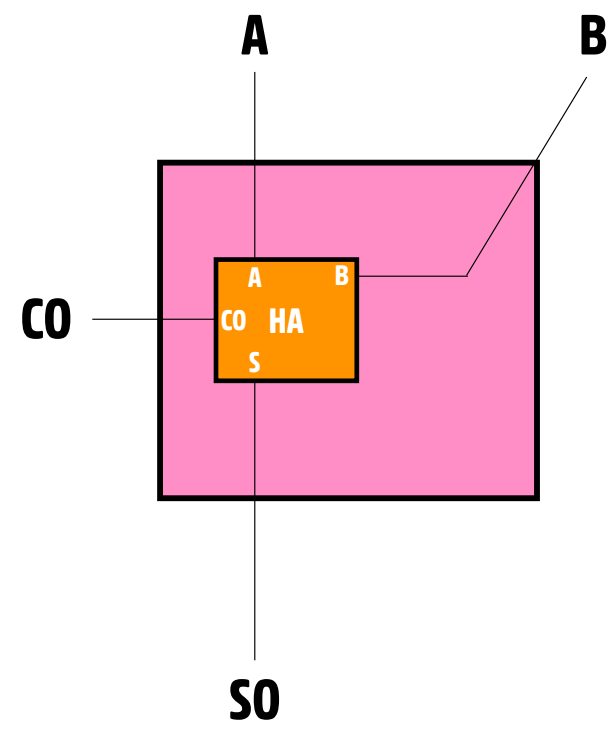
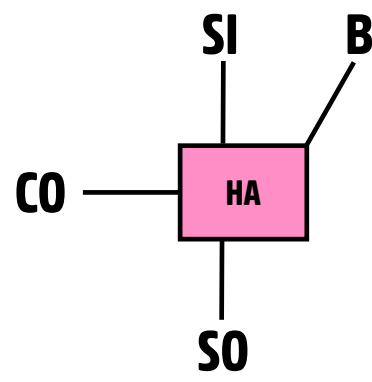
Binary Multiplication - Block #3



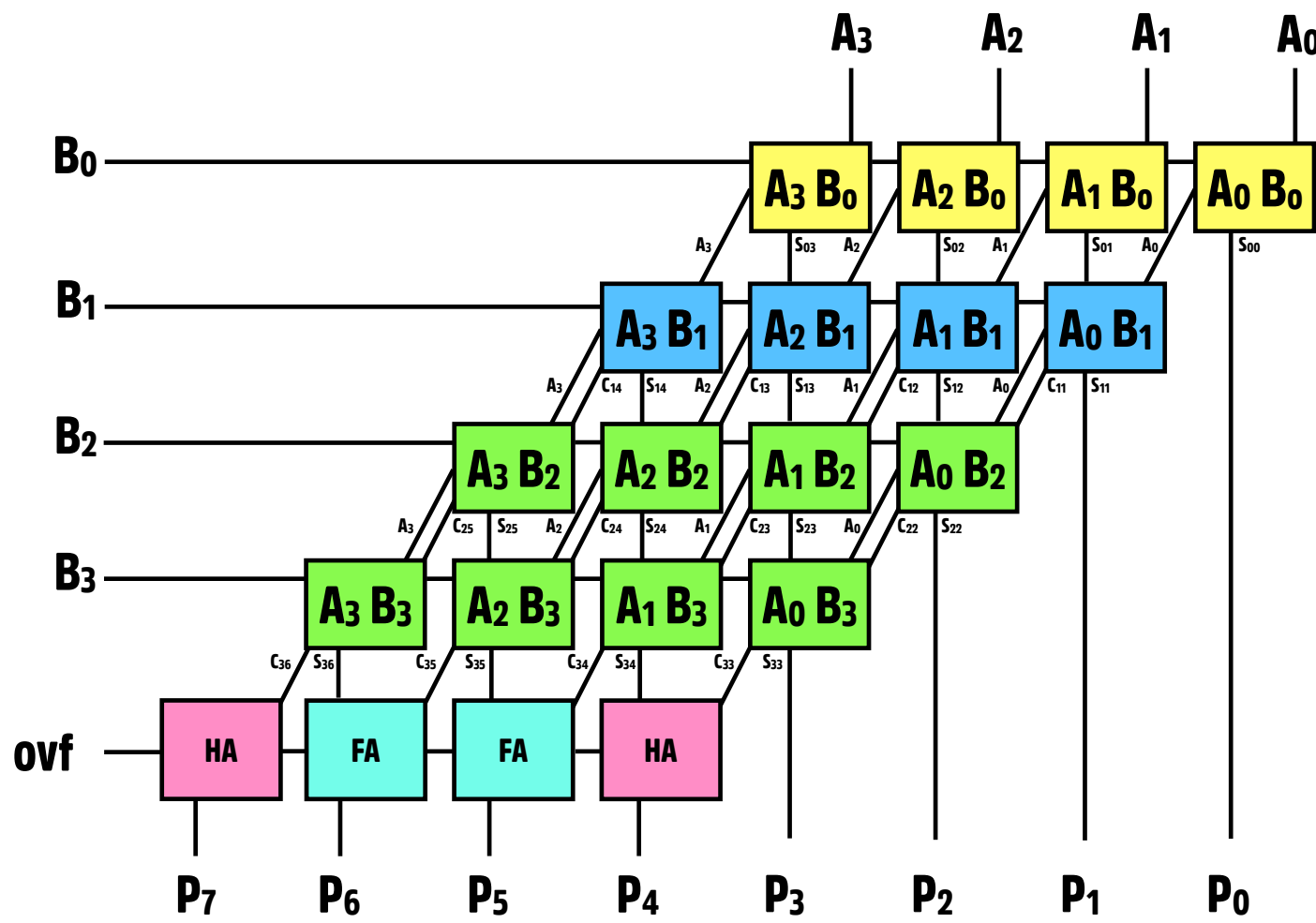
Binary Multiplication - Block #4



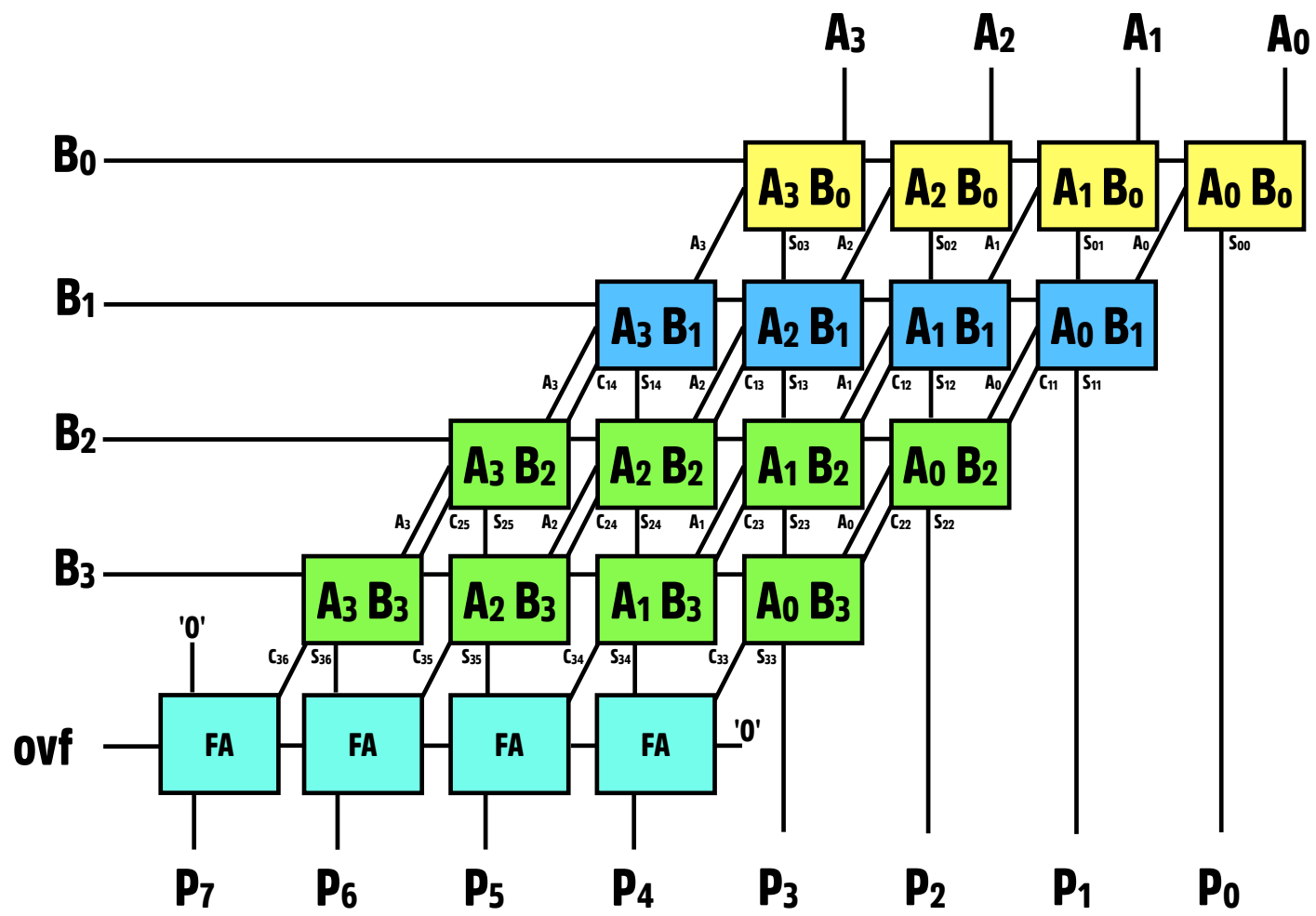
Binary Multiplication - Block #5



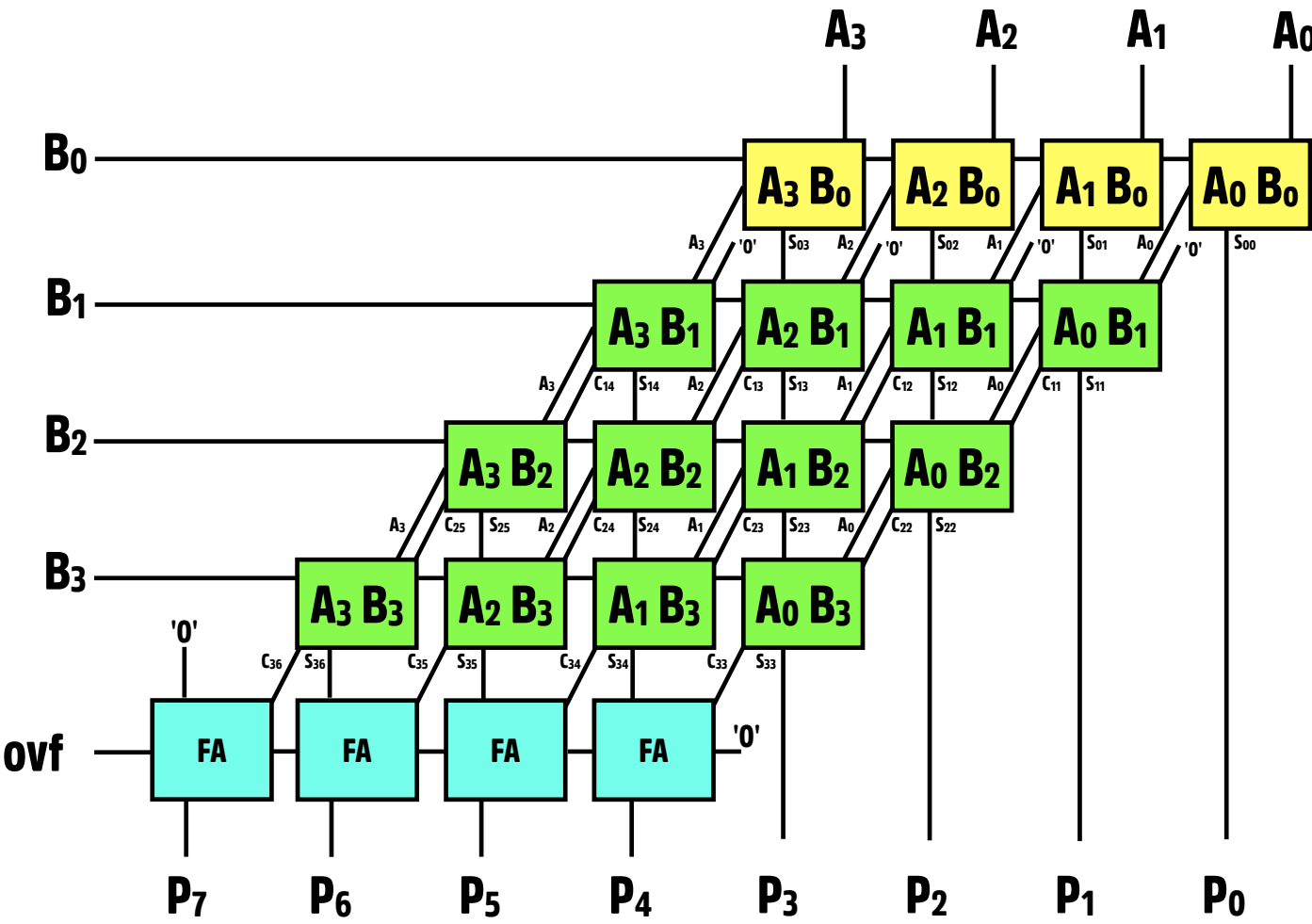
Binary Multiplier - Version #1



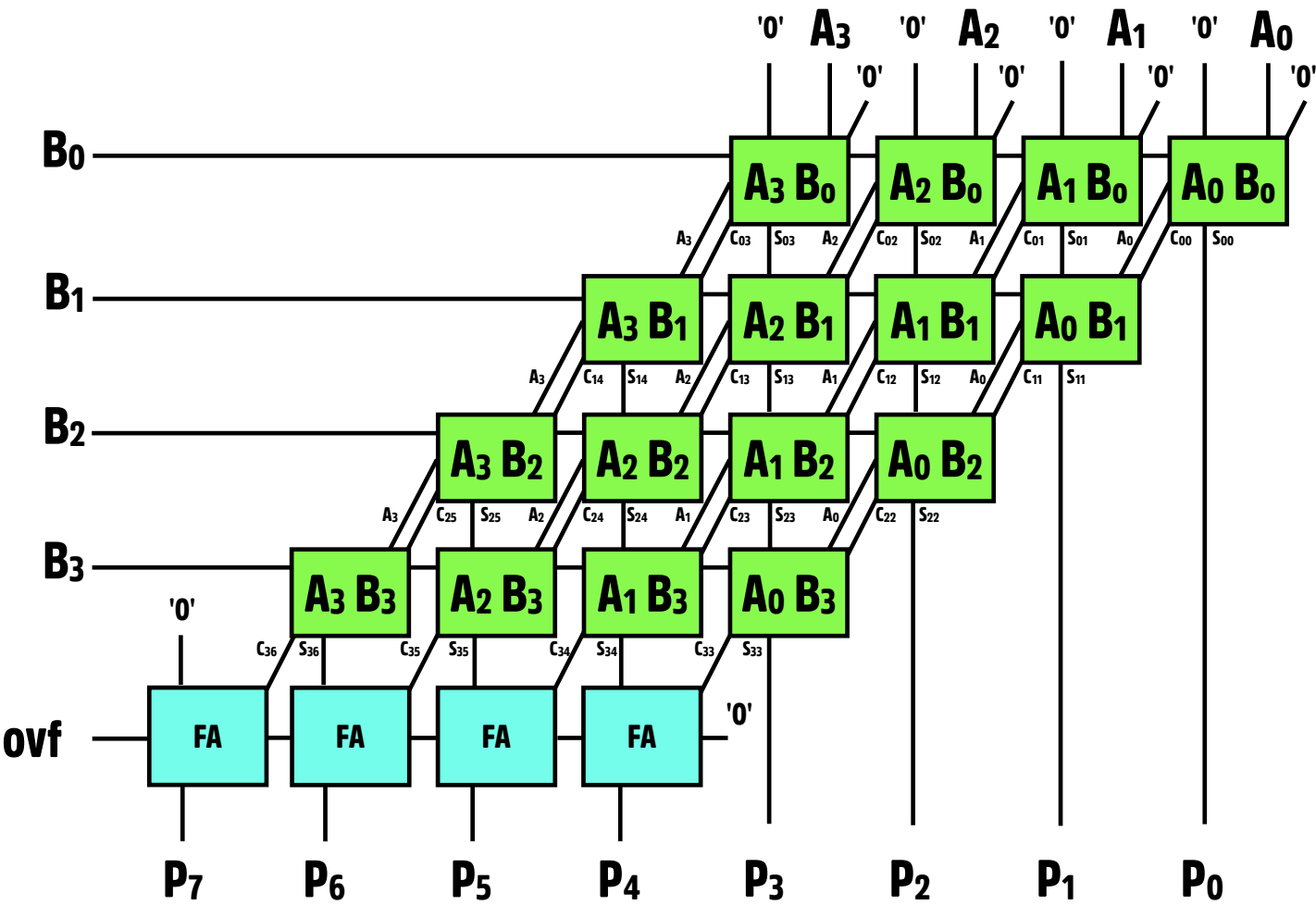
Binary Multiplier - Version #2



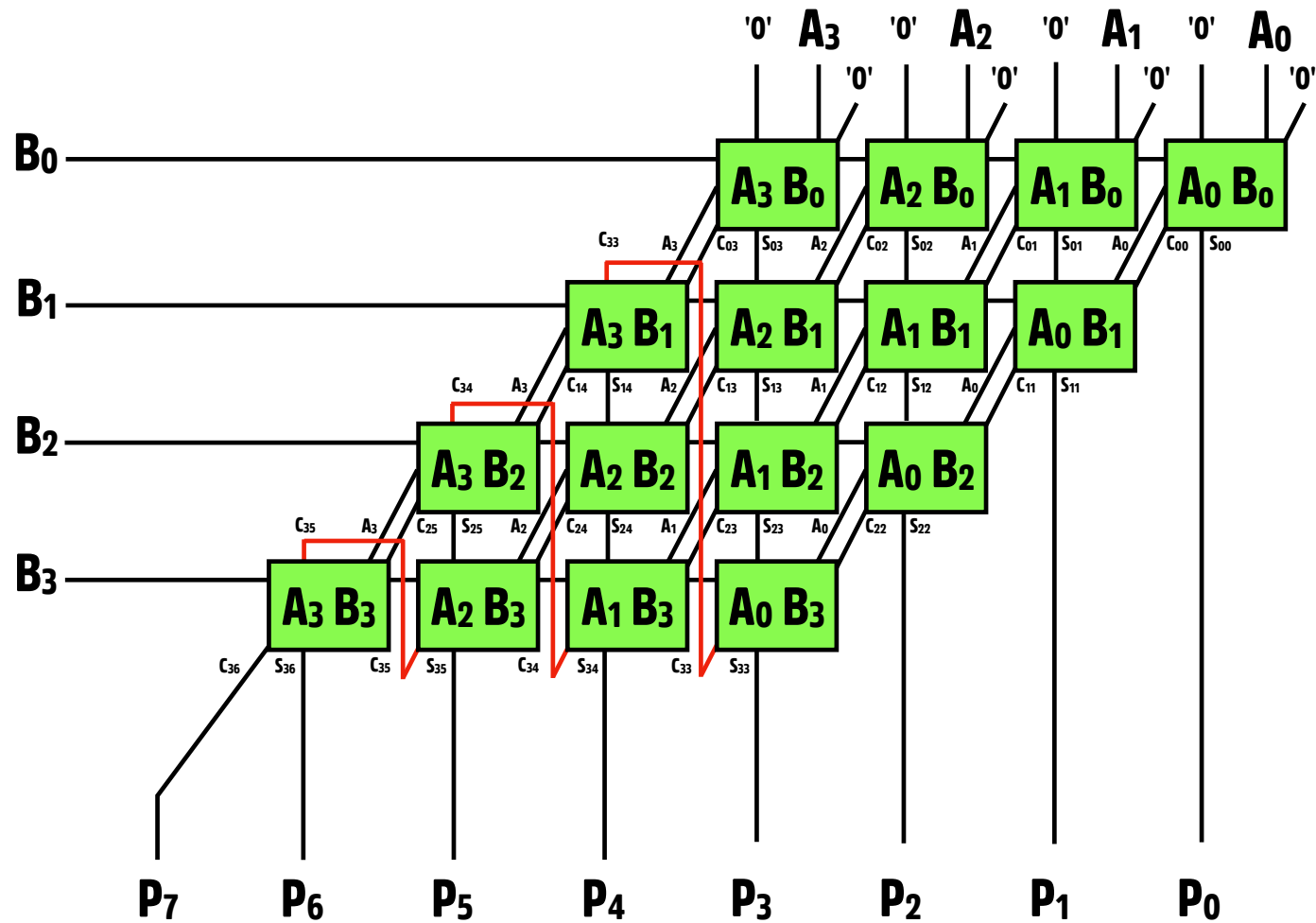
Binary Multiplier - Version #3



Binary Multiplier - Version #4



Binary Multiplier - Version #5



END